

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

Dmitry Kurochkin

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Lecture 10

1 Instrumental Variables (IV)

- IV Estimator with a Single Regressor or and a Single instrument
 - Terminology and Assumptions
 - Two-stage Least Squares (2SLS)
 - Sampling Distribution of $\hat{\beta}_1^{2SLS}$
- General IV Regression Model
 - Terminology and Assumptions
 - Instrumental Variables in General IV Regression Model
 - 2SLS in General IV Model
 - The Two Conditions for Valid Instruments
 - Sampling Distribution of 2SLS Estimators in General IV Model
- R Example
 - Simulation
 - 2SLS via 'ivreg'

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IV Estimator with a Single Regressor or and a Single instrument

Terminology:

Assume that the (true) population model is

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n,$$

where u_i is the error term.

- If $\text{Cov}(X_i, u_i) \neq 0$ (e.g., missing variables or simultaneous causality), then the OLS estimator is biased. X_i in such cases is called *endogenous* variable.
- Z_i is called *instrumental variable* (or *instrument*) if the following two conditions hold:
 - 1 *Instrument relevance*: $\text{Cov}(Z_i, X_i) \neq 0$.
 - 2 *Instrument exogeneity*: $\text{Cov}(Z_i, u_i) = 0$.

Two-stage Least Squares (2SLS)

Assume that the (true) population model is

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n,$$

and let Z_i be an IV, that is, assume that the following two conditions hold:

- ① *Instrument relevance*: $\text{Cov}(Z_i, X_i) \neq 0$.
- ② *Instrument exogeneity*: $\text{Cov}(Z_i, u_i) = 0$.

Then an unbiased estimate of β_1 can be obtained in two stages as follows.

Two-stage Least Squares (2SLS):

- Stage 1:
Regress X_i on Z_i via $X_i = \pi_0 + \pi_1 Z_i + v_i$ to get $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i$.
- Stage 2:
Regress Y_i on $\hat{X}_i$ to get the 2SLS estimator $\hat{\beta}_1^{2SLS}$ of β_1 .

Sampling Distribution of $\hat{\beta}_1^{2SLS}$

Claim:

Given IV regression,

$$\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}.$$

Claim:

Given IV regression, the 2SLS estimator is given by:

$$\hat{\beta}_1^{2SLS} = \frac{S_{ZY}}{S_{ZX}},$$

where

- S_{ZY} is the sample covariance between Z_i and Y_i ;
- S_{ZX} is the sample covariance between Z_i and X_i .

Sampling Distribution of $\hat{\beta}_1^{2SLS}$

Claim:

Given IV regression, the 2SLS estimator is consistent:

$$\hat{\beta}_1^{2SLS} = \frac{S_{ZY}}{S_{ZX}} \xrightarrow{P} \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)} = \beta_1,$$

as $n \rightarrow \infty$.

Claim:

Given IV regression, the 2SLS estimator is approximately normal, in large samples:

$$\hat{\beta}_1^{2SLS} \xrightarrow{D} N(\beta_1, \sigma_{\hat{\beta}_1^{2SLS}}^2) \text{ as } n \rightarrow \infty, \text{ where } \sigma_{\hat{\beta}_1^{2SLS}}^2 = \frac{1}{n} \frac{\text{Var}((Z_i - \mu_Z)u_i)}{(\text{Cov}(Z_i, X_i))^2}.$$

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General IV Regression Model

Terminology:

Assume that the (true) population model is

$$Y_i = \beta_0 + \underbrace{\beta_1 X_{i1} + \dots + \beta_k X_{ik}}_{\text{endogenous regressors}} + \underbrace{\beta_{k+1} W_{i1} + \dots + \beta_{k+r} W_{ir}}_{\text{exogenous regressors}} + u_i,$$

where

- index $i \in \{1, 2, \dots, n\}$ and the samples are drawn randomly
- u_i is the error term
- Y_i is the dependent variable
- $\beta_0, \beta_1, \dots, \beta_{k+1}$ are unknown regression coefficients
- $X_{i1}, \dots, X_{ik}$ are k endogenous (“problematic”) regressors – they can correlate with u_i
- $W_{i1}, \dots, W_{ir}$ are r exogenous regressors: $E[u_i | W_{i1}, \dots, W_{ir}] = 0$ for all j

Instrumental Variables in General IV Regression Model

Instrumental Variables (IV):

Given

$$Y_i = \beta_0 + \underbrace{\beta_1 X_{i1} + \dots + \beta_k X_{ik}}_{\text{endogenous regressors}} + \underbrace{\beta_{k+1} W_{1i} + \dots + \beta_{k+r} W_{ri}}_{\text{exogenous regressors}} + u_i,$$

population model,

$Z_{i1}, \dots, Z_{mi}$ are called *instrumental variables* (or *instruments*) if

$$\text{Cov}(Z_{ij}, u_i) = 0, \quad \text{for all } j \in \{1, \dots, m\}.$$

This condition is called *instrument exogeneity*.

Instrumental Variables in General IV Regression Model

Given k endogenous regressors X_{ij} and m instrumental variables Z_{ij} , the regression coefficients

$$\beta_0, \beta_1, \dots, \beta_{k+r}$$

are said to be

- *exactly identified* if $m = k$
- *overidentified* if $m > k$ (more instrument)
- *underidentified* if $m < k$ (less instrument)

Note: To get unbiased estimators, one must have either $m = k$ (exactly identified case) or $m > k$ (overidentified case).

2SLS in General IV Model

First, assume that there is one endogenous (“problematic”) regressor X_i :

$$Y_i = \beta_0 + \underbrace{\beta_1 X_i}_{\text{endogenous regressor}} + \underbrace{\beta_2 W_{1i} + \dots + \beta_{k+r} W_{ri}}_{\text{exogenous regressors}} + u_i,$$

and let $Z_{i1}, \dots, Z_{im}$ be IVs, that is, assume that $\text{Cov}(Z_{ij}, u_i) = 0$ for all $j \in \{1, \dots, m\}$.

Then unbiased estimates of β 's can be obtained in two stages as follows.

Two-stage Least Squares (2SLS):

- Stage 1:

Regress X_i on instruments Z 's and exogenous variables W 's via

$$X_i = \pi_0 + \pi_1 Z_{i1} + \dots + \pi_m Z_{im} + \pi_{m+1} W_{i1} + \dots + \pi_{m+r} W_{ir} + v_i$$

to get $\hat{X}_i$.

- Stage 2:

Regress Y_i on $\hat{X}_i$ and $W_{i1}, \dots, W_{ir}$ to get the 2SLS estimators of $\beta_0, \beta_1, \dots, \beta_{k+r}$.

2SLS in General IV Model

In a more general case, there can be k endogenous regressors X 's:

$$Y_i = \beta_0 + \underbrace{\beta_1 X_{i1} + \dots + \beta_k X_{ik}}_{\text{endogenous regressors}} + \underbrace{\beta_{k+1} W_{i1} + \dots + \beta_{k+r} W_{ir}}_{\text{exogenous regressors}} + u_i.$$

The estimates of β 's can be obtained in two stages as follows.

Two-stage Least Squares (2SLS):

- Stage 1:

Regress each X_{ij} on instruments Z 's and exogenous variables W 's via

$$X_{i1} = \pi_{01} + \pi_{11} Z_i + \dots + \pi_{m1} Z_{im} + \pi_{m+1,1} W_{i1} + \dots + \pi_{m+r,1} W_{ir} + v_{i1}$$

$$\vdots$$

$$X_{ik} = \pi_{0k} + \pi_{1k} Z_i + \dots + \pi_{mk} Z_{im} + \pi_{m+1,k} W_{i1} + \dots + \pi_{m+r,k} W_{ir} + v_{ik}$$

to get $\hat{X}_{i1}, \dots, \hat{X}_{ik}$.

- Stage 2:

Regress Y_i on $\hat{X}_{i1}, \dots, \hat{X}_{ik}$ and $W_{i1}, \dots, W_{ir}$ to get the 2SLS estimators of $\beta_0, \beta_1, \dots, \beta_{k+r}$.

The Two Conditions for Valid Instruments

Let's summarize the two conditions for instruments in the General IV Regression case:

- ① *Instrument relevance*: $1, \hat{X}_{i1}, \dots, \hat{X}_{ik}, W_{i1}, \dots, W_{ir}$ are not perfectly collinear.
- ② *Instrument exogeneity*: $\text{Cov}(Z_{ij}, u_i) = 0$ for all $j \in \{1, \dots, m\}$.

Sampling Distribution of 2SLS Estimators in General IV Model

Claim:

Given

$$Y_i = \beta_0 + \underbrace{\beta_1 X_{i1} + \dots + \beta_k X_{ik}}_{\text{endogenous regressors}} + \underbrace{\beta_{k+1} W_{1i} + \dots + \beta_{k+r} W_{ri}}_{\text{exogenous regressors}} + u_i.$$

population model with assumptions defined above. If also X 's, W 's, Y 's, and Z 's have nonzero finite fourth moments, then

- 2SLS estimators are consistent
- 2SLS estimators are approximately normal, as $n \rightarrow \infty$

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R Example: Simulation

```
cause <- function(A, B = NULL, C = NULL, D = NULL) {  
  if(!is.null(B)){ A = A + B }  
  if(!is.null(C)){ A = A + C }  
  if(!is.null(D)){ A = A + D }  
  u = rnorm(length(A), sd = 0.1)  
  return(A + u)  
}
```

```
U = runif(10000)  
Z = runif(10000)  
X1 = cause(Z)  
X2 = cause(U)  
X = 0.7*X1 + 0.3*X2  
Y = cause(X,U)
```

```
fit0 <- lm(Y ~ X + U)  
fit1 <- lm(Y ~ X)
```

```
X_hat <- predict(lm(X ~ Z))  
fit2 <- lm(Y ~ X_hat)
```

R Example: Simulation

```
stargazer(fit0, fit1, fit2, type = "text")
```

Dependent variable:			
	(1)	Y (2)	(3)
X	0.997*** (0.005)	1.468*** (0.012)	
U	1.003*** (0.004)		
X_hat			0.998*** (0.020)
Constant	0.001 (0.003)	0.268*** (0.007)	0.501*** (0.011)
Observations	10,000	10,000	10,000
R2	0.952	0.586	0.204
Adjusted R2	0.952	0.586	0.204
Residual Std. Error	0.098 (df = 9997)	0.286 (df = 9998)	0.397 (df = 9998)
F Statistic	98,272.050*** (df = 2; 9997)	14,169.150*** (df = 1; 9998)	2,562.672*** (df = 1; 9998)
Note: *p<0.1; **p<0.05; ***p<0.01			

R Example: 2SLS via 'ivreg'

```
library("ivreg")  
fit <- ivreg(Y ~ X | Z)  
summary(fit)
```

Call:
ivreg(formula = Y ~ X | Z)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.7736849	-0.2500600	0.0002677	0.2499492	0.8100239

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.501235	0.008161	61.42	<2e-16 ***
X	0.998224	0.015212	65.62	<2e-16 ***

Residual standard error: 0.3063 on 9998 degrees of freedom

Multiple R-Squared: 0.5263, Adjusted R-squared: 0.5263

Wald test: 4306 on 1 and 9998 DF, p-value: < 2.2e-16